CNENGR115 Computers and Control

First Order Systems (1)

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Control Lectures

Part 1 – Introduction

Part 2 – Mathematical Models

Part 3 – Control Actions

What is a Mathematical Model?

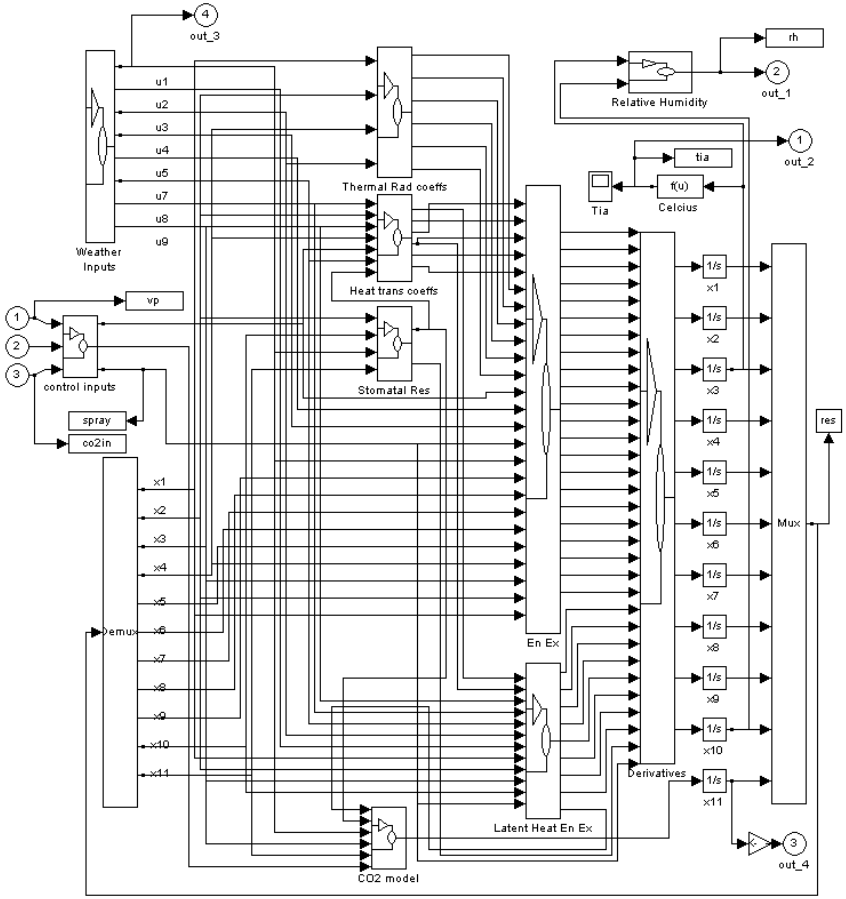
"A scientific hypothesis about the behaviour of the system under study"

Why model?

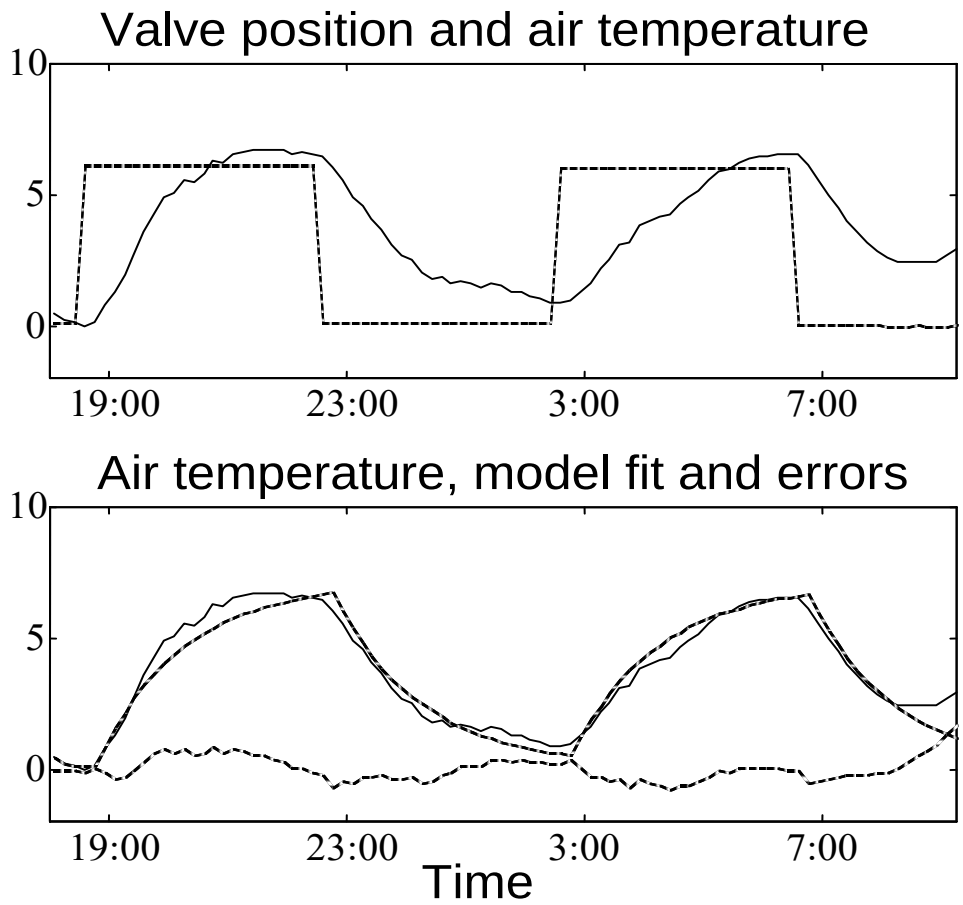
- Scientific understanding
- Forecasting
- Simulation
- Management / operational planning
- Control system design

Agricultural Building

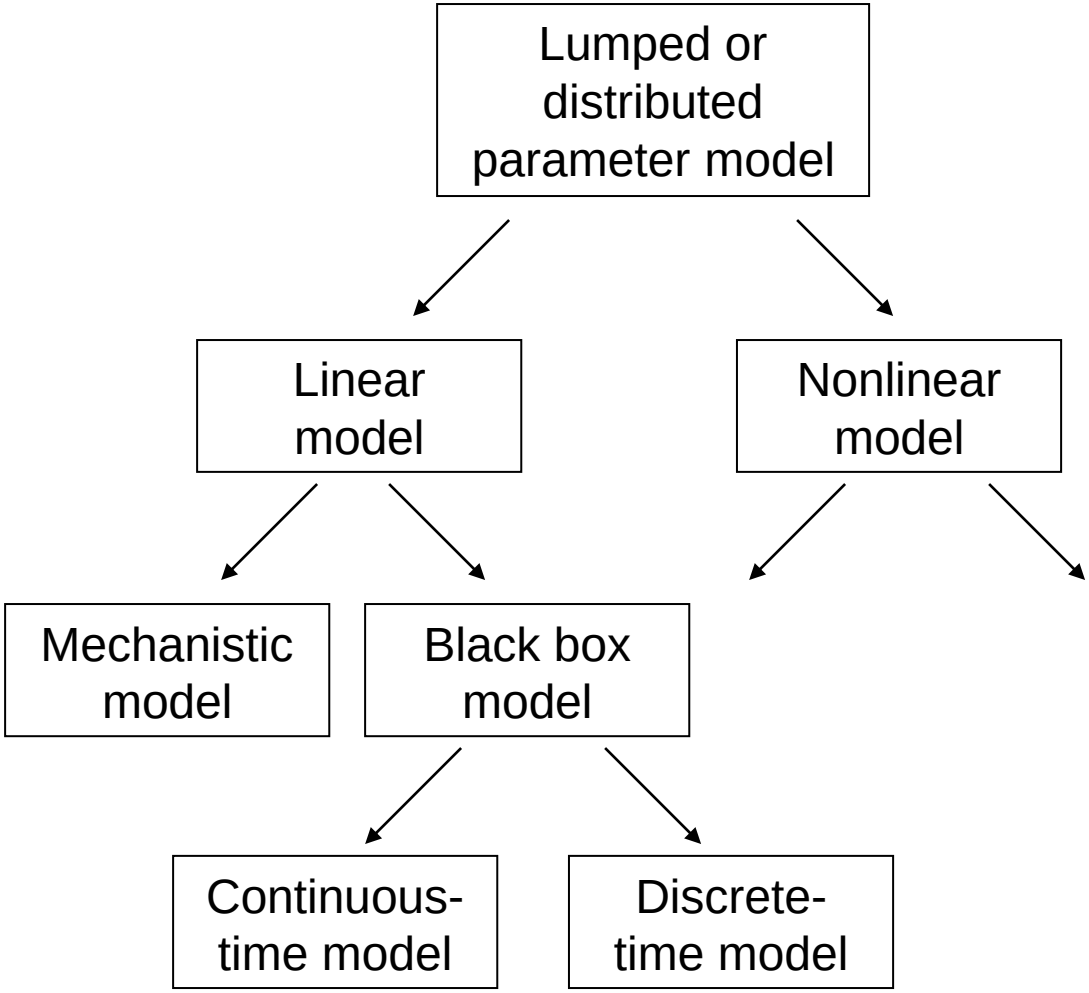
Matlab / Simulink Glasshouse Model



Compare Model Output with Real Data



Types of Model – Tree Diagram



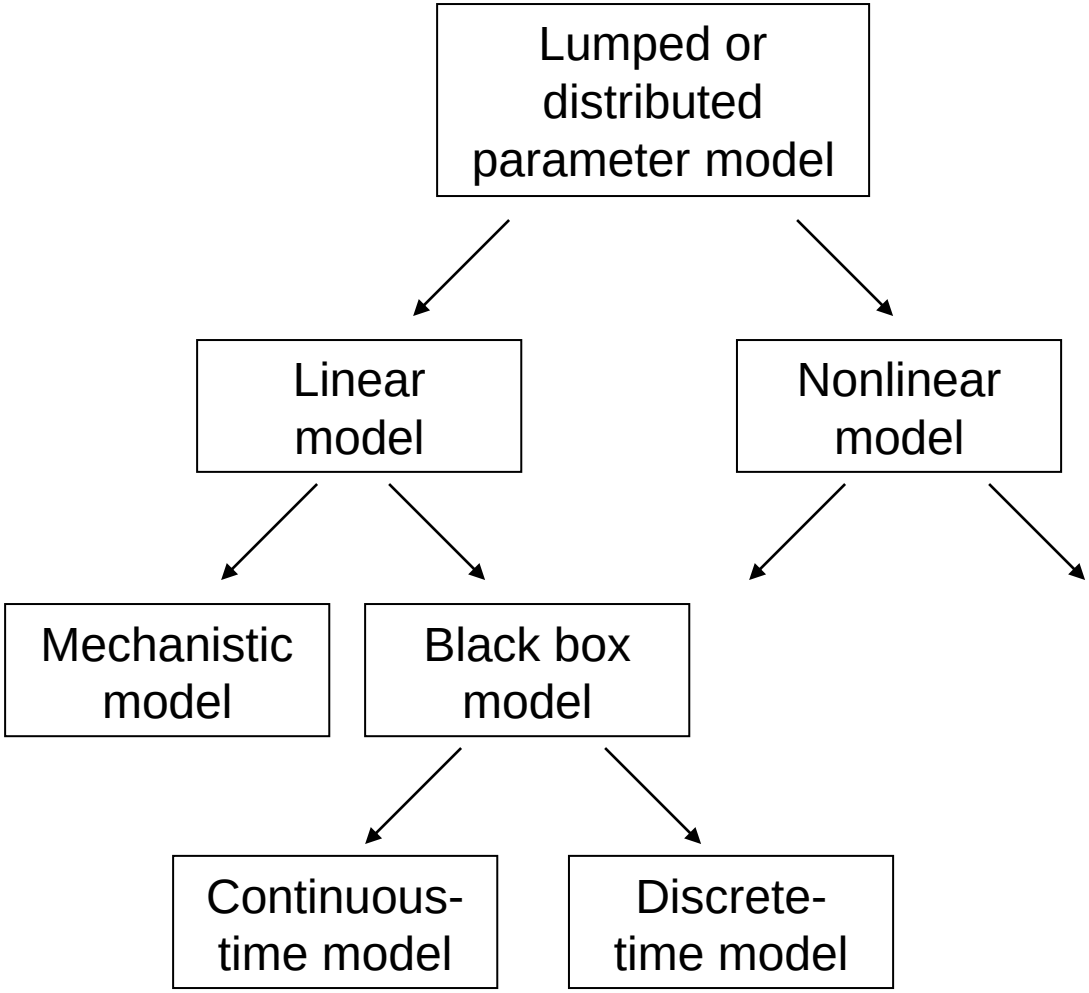
Distributed Parameter Models

- Partial differential equations describing how the variables change over all points of time and space, e.g. **Computational Fluid Dynamics** (CFD) model
- The CFD model for a ventilation system attempts to describe variations over time of the velocity (speed and direction) of airflow within the entire 3-D space

Lumped Parameter Models

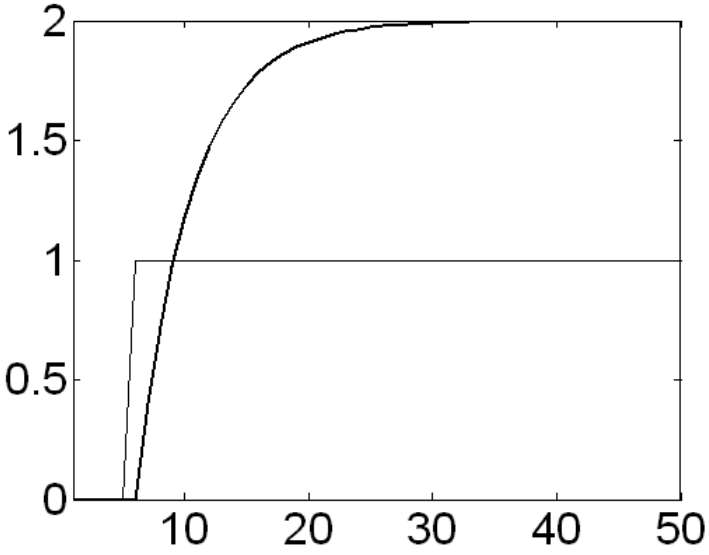
- Ordinary differential equations describing how the variables change in time between **fixed points in space**
- Example – Transfer Function model (see later lecture) describing the relationship between airflow at the inlet and outlet of a ventilation system

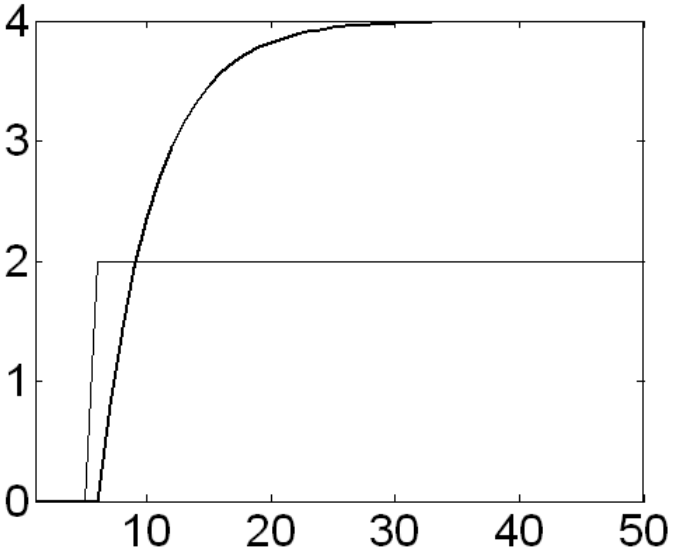
Types of Model – Tree Diagram



Linear Models

- Obey the **principle of superposition**
- Stability is an **inherent** property of a linear system, which is independent of the nature and amplitude of any external inputs





Nonlinear Models

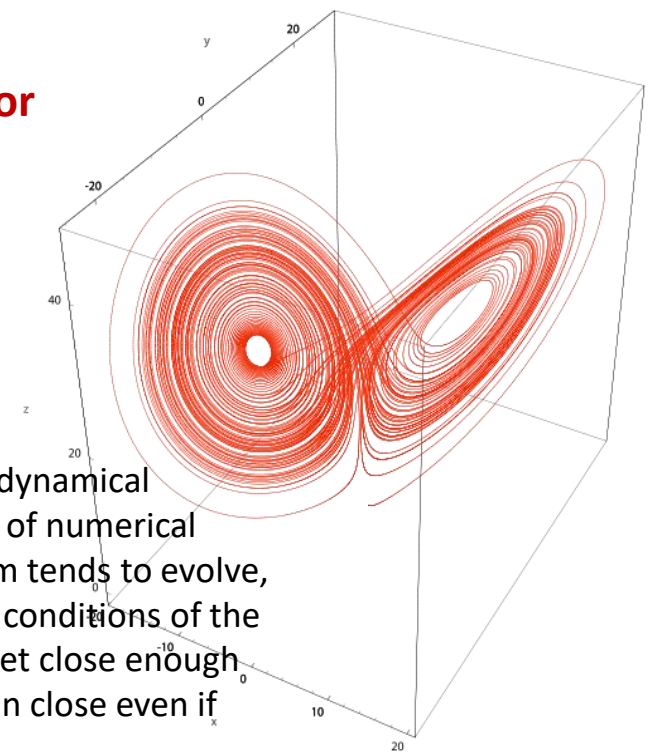
- Do **not** obey the principle of superposition and can exhibit quite different response to different sized but similarly shaped inputs
- Can exhibit complex or even **chaotic behaviour**

$$\frac{dx}{dy} = \sigma(y - x)$$

$$\frac{dy}{dt} = x(p - z) - y$$

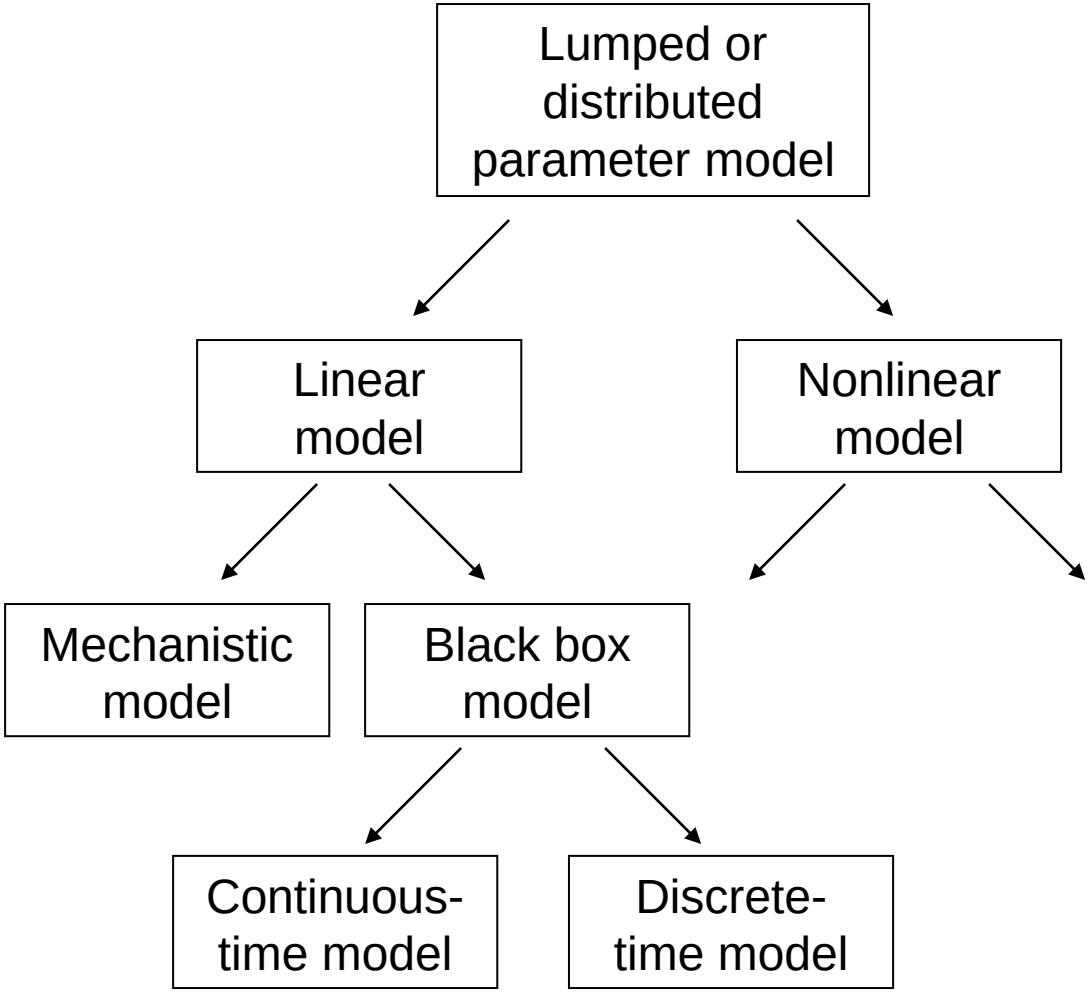
$$\frac{dz}{dy} = xy - \beta z$$

Lorenz attractor



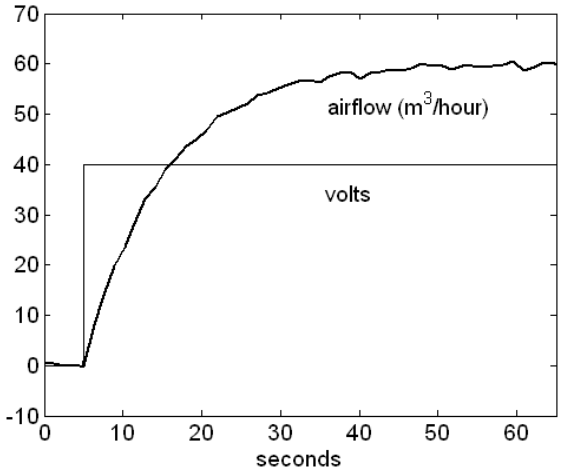
In the mathematical field of dynamical systems, an **attractor** is a set of numerical values toward which a system tends to evolve, for a wide variety of starting conditions of the system. System values that get close enough to the **attractor** values remain close even if slightly disturbed.

Types of Model – Tree Diagram



Black Box Models

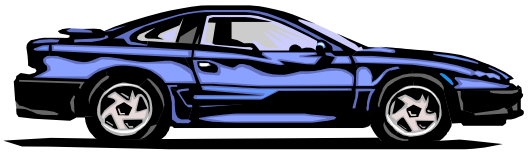
- Relate the input and output variables in the simplest possible manner, without regard to internal mechanisms
- Parameters are typically estimated from data using **graphical** or **statistical** methods



Mechanistic Models

- Usually obey known **physical laws**, as perceived by the model builder or scientist
- For example, we will use **Newton's Law** for translational motion to develop a mechanistic model for vehicle speed

$$\tau \dot{x} + x = Ku(t)$$



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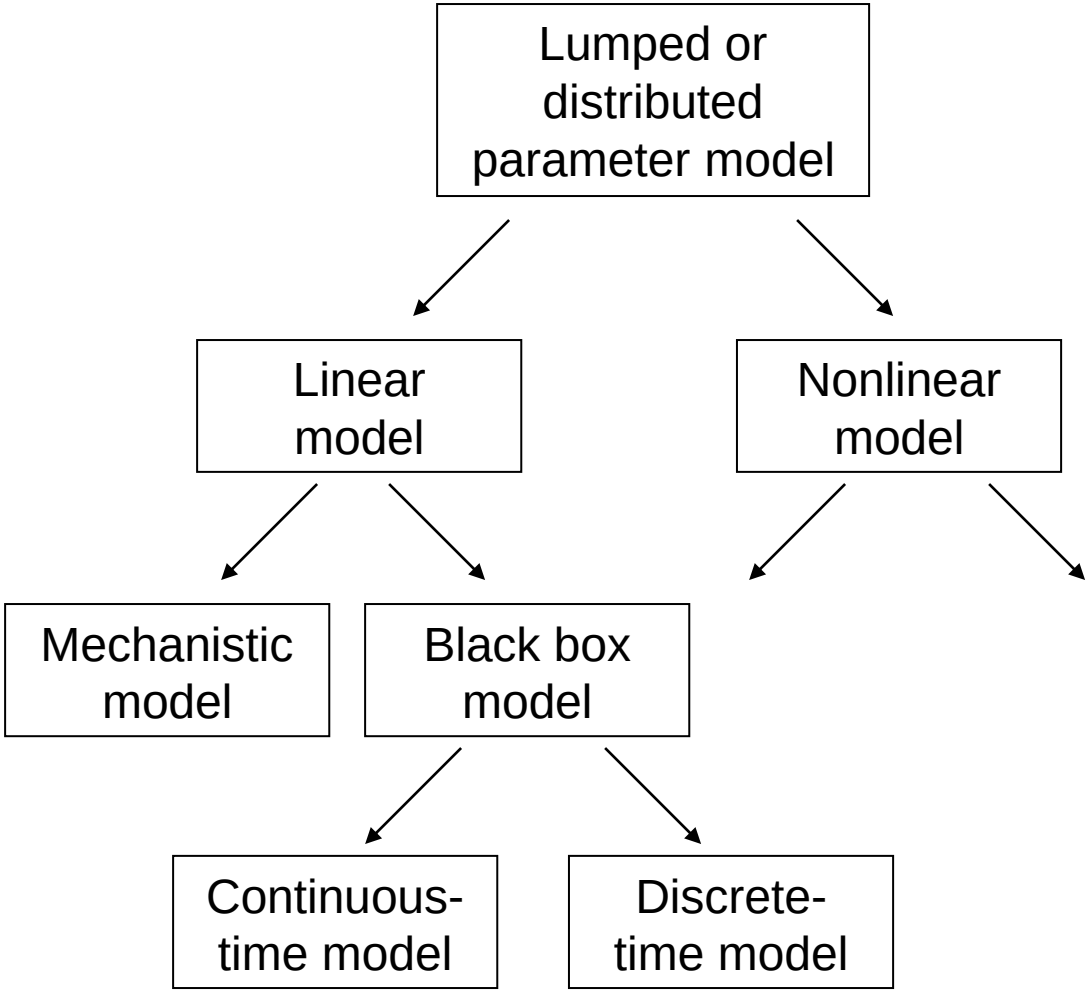
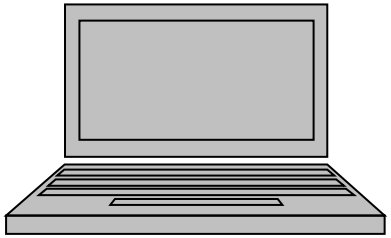


Continuous–Time Models

- Systems that vary continuously are called continuous–time or **analogue** systems
- They are typically represented using differential equations

Discrete–Time Models

- Systems that vary at discrete intervals in time are called **digital** systems
- They can be modelled using **difference equations**
- A continuous time system is often controlled using a digital computer by **sampling** the output at regular intervals



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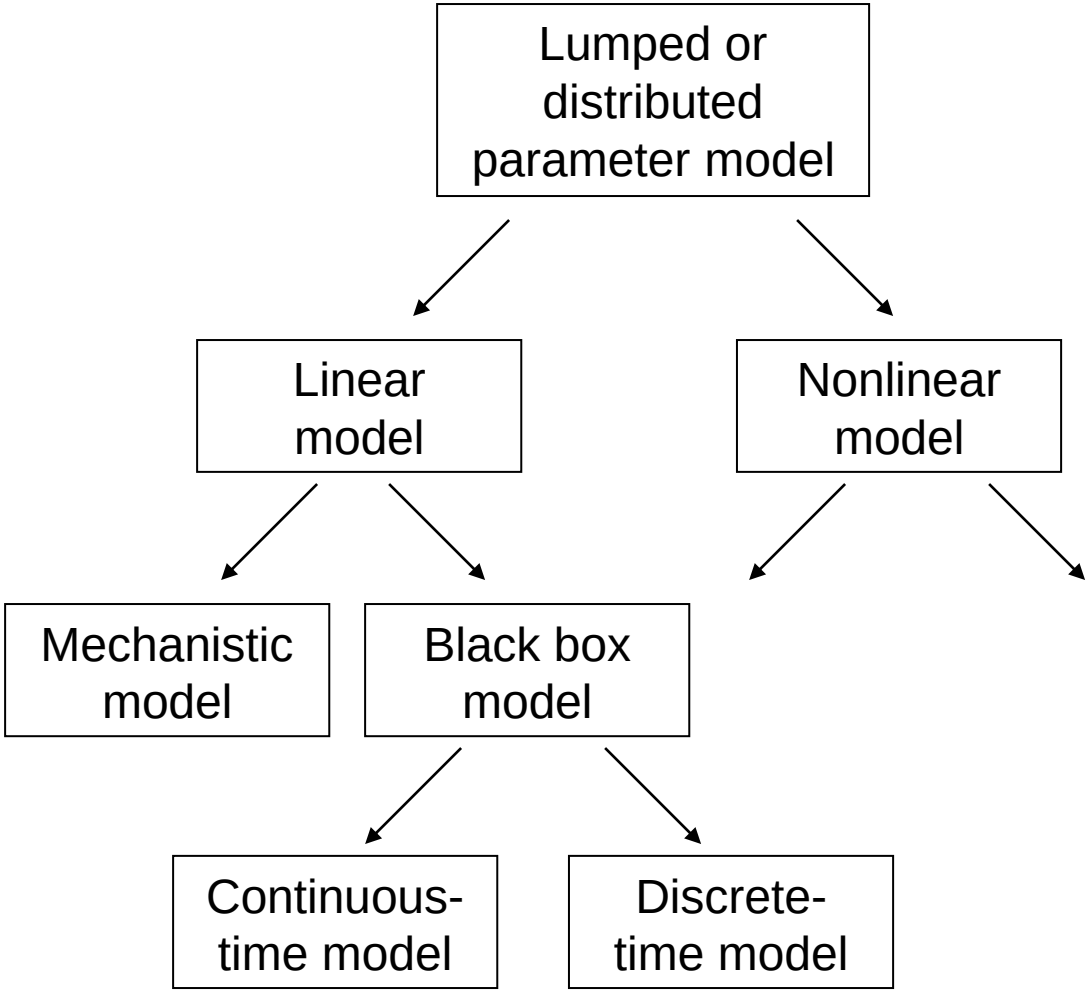


Exercise

A dynamic model represents the behaviour of something over time.

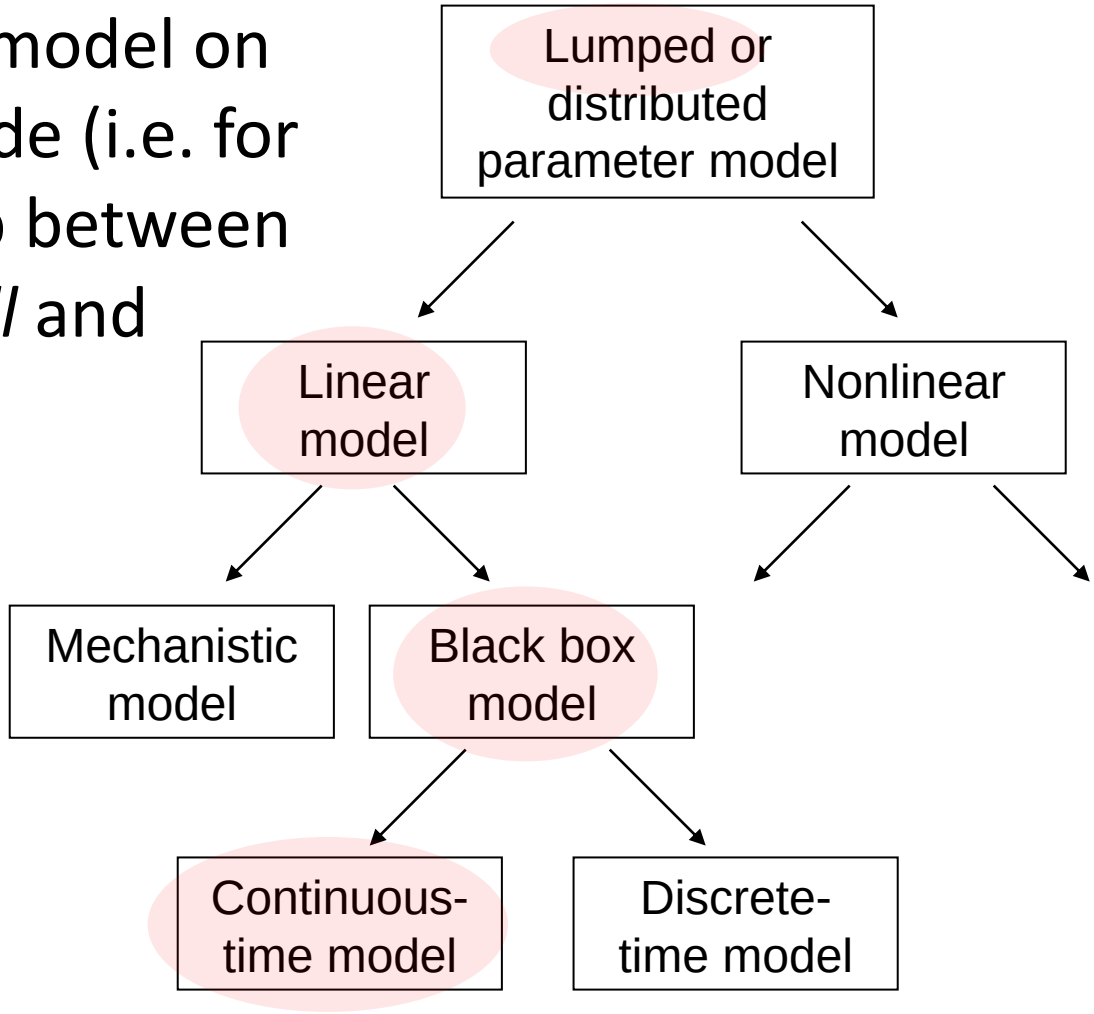
Exercise

$$y(t) = \frac{0.01706s^2 + 0.1578s + 0.001680}{s^2 + 0.3299s + 0.001768} u(t)$$

- $y(t)$ is the **river flow** output
- $u(t)$ is the "**effective rainfall**" input
- s is the differential operator, i.e. d/dt
- Obtained by statistical estimation from the hourly measured flow and effective rainfall

Exercise

Categorise the model on the previous slide (i.e. for the relationship between *effective rainfall* and river flow)



Data-Based Mechanistic (DBM) models

- Data are used to identify both the model structure and to estimate the parameters using **statistical analysis**
- But the DBM model is only credible if it is also capable of **interpretation** in a physically meaningful manner

- e.g. Price, L., Young, P., Berckmans, D., Janssens, K. and Taylor, J. (1999) Data–Based Mechanistic Modelling (DBM) and control of mass and energy transfer in agricultural buildings, Annual Reviews in Control, **23**, pp. 71–82.

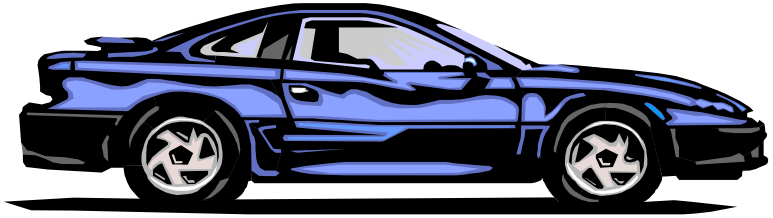
Data-Based Mechanistic (DBM) models

- Applying partial fraction expansion, the river flow model can be written in parallel pathway form:

$$y(t) = \left\{ 0.0171 + \frac{0.1496}{s + 0.3244} + \frac{0.002573}{s + 0.00545} \right\} u(t)$$

- This has the following practical interpretation:
  - an **instantaneous effect** accounting for the small amount of effective rainfall that affects the river flow within 1 hour
  - 'quick–flow' effect with steady state gain and residence time or time constant (see later lecture) that is associated with **surface and near surface processes**
  - a 'slow–flow' effect with steady state gain and residence associated with **groundwater processes**

Mathematical Model for Vehicle Speed

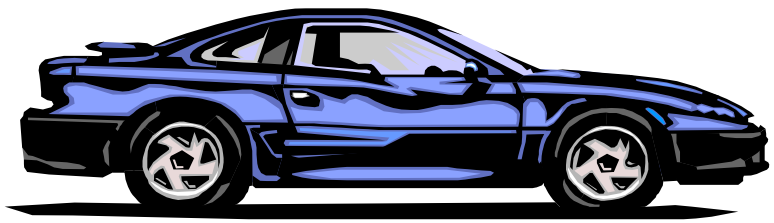


- To be able to understand, analyse and then control dynamic systems, we can model them mathematically
- We will develop a linear continuous-time, mechanistic model for the speed of a vehicle
- Denote the **displacement** of the vehicle from a given reference point by y (m)
- The rate of change of displacement is the **velocity** $\dot{y}$ (m/s)
- The rate of change of the velocity is the **acceleration** $\ddot{y}$ (m/s^2)

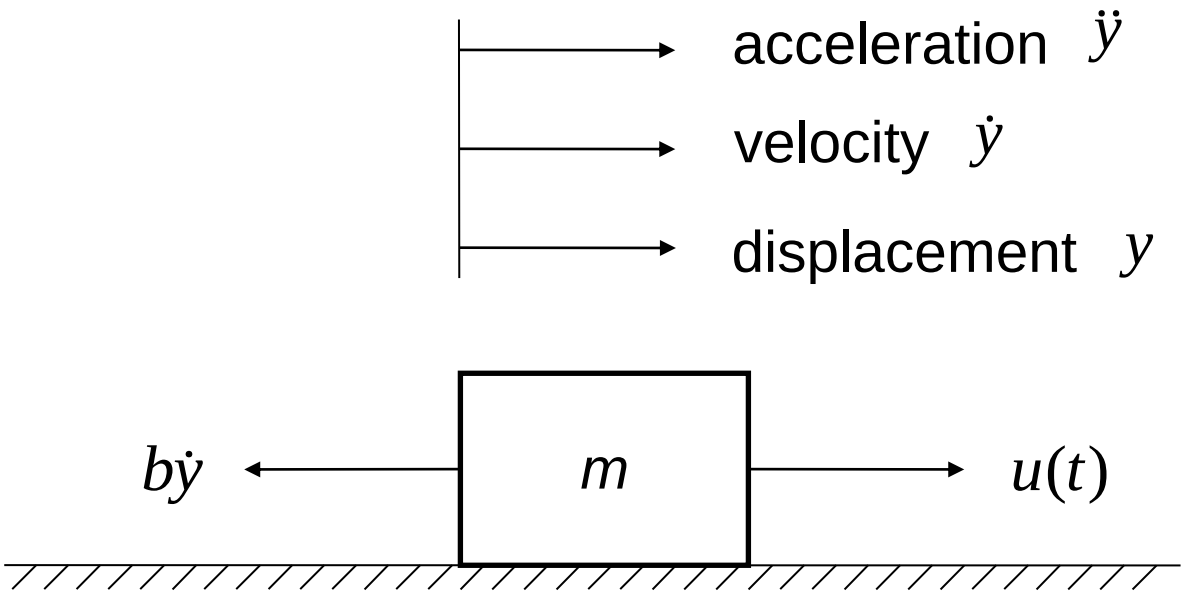
Displacement – Speed – Acceleration

$$y \longrightarrow \dot{y} \longrightarrow \ddot{y}$$

$$v \longrightarrow \dot{v}$$



Free Body Diagram for Vehicle Speed



- Assume the vehicle has a **mass** m and the engine imparts a **force** $u(t)$
- Assume that the rotational inertia of the wheels is negligible
- Assume that **friction** slows the car down and is proportional to the velocity, i.e. friction force equals $b\dot{y}$ where b is a coefficient

Mathematical Model for Vehicle Speed

- Using Newton's Law (force equals mass x acceleration),

$$u(t) - b\dot{y} = m\ddot{y}$$

- We will treat engine force as an input and the displacement as an output

$$m\ddot{y} + b\dot{y} = u(t)$$

- This is a **2nd order** differential equation
- Later in the course we will consider a cruise control system that automatically adjusts engine force to regulate the speed
- In this case, we are primarily interested in the speed

$$m\dot{v} + bv = u(t)$$

- This is a **1st order** differential equation

Standard Notation for a 1st Order System

$$\tau \dot{x} + x = Ku(t) \quad \text{OR} \quad \boldsymbol{\tau} \frac{dx}{dt} + x = Ku(t)$$

output variable x

input variable $u(t)$

time constant τ

steady state gain K

Analysis of Vehicle Speed Model

- This is our model for vehicle speed

$$m\dot{v} + bv = u(t)$$

- By convention, the model is often re-arranged as follows

$$\frac{m}{b}\dot{v} + v = \frac{1}{b}u(t)$$

- Compare with the general 1st order model

$$\tau\dot{x} + x = Ku(t)$$

Output

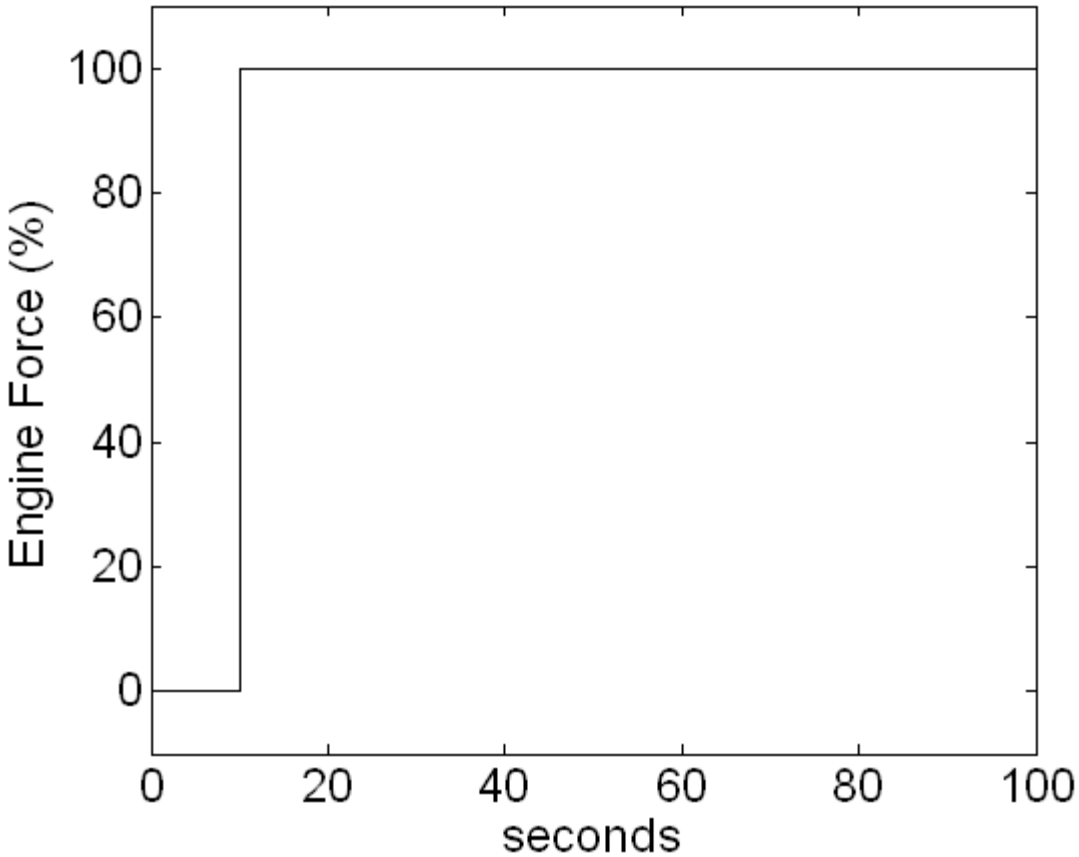
Input

Time constant

Steady state gain

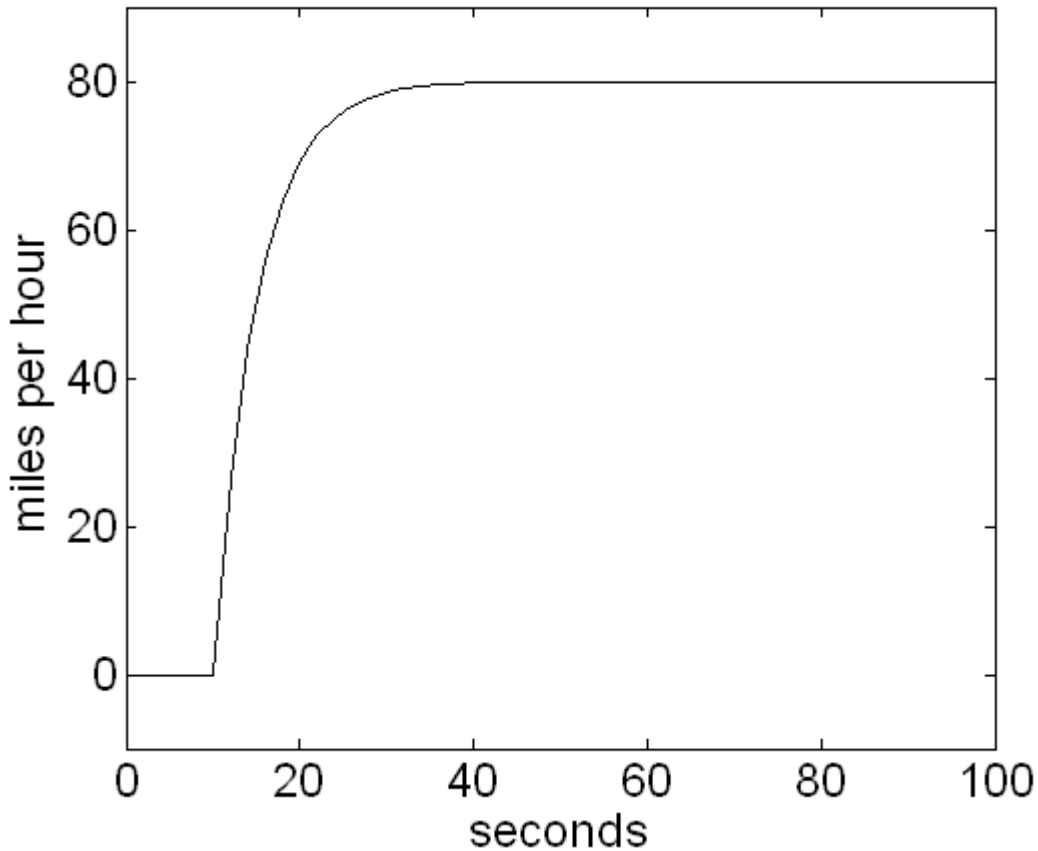
Engine Force Step Input

- In this experiment, the engine force is a **step input** from zero to 100%
- Assume that 0% is zero force and that 100% represents full power



Vehicle Speed Step Response

- Given typical values for mass and friction, the response of the speed to a 100% step in engine power is shown (right)
- A **simulation package** was used to solve the differential equation and obtain this graph (Matlab/Simulink)



Control Lectures

Part 1 – Introduction

Part 2 – Mathematical Models

Part 3 – Control Actions